

FlowTrace Deployment Guide (Under 20 Mins)

Deploy the full-stack FlowTrace application using Vercel, Render, and MongoDB Atlas.

Fast-Track Deployment Requirement: Your MongoDB Atlas connection is already active and tested successfully! You only need to host the React client and the Express backend.

1. Create your Git Repository

Push your local **flowtrace** folder to a private or public GitHub/GitLab repository. Both Render and Vercel will build directly from your Git commits.

2. Deploy Frontend on Vercel (Time: 3 Mins)

Vercel hosts the React + Vite static client.

1. Sign up/log in at **Vercel.com**.
2. Click **Add New > Project**.
3. Import your **flowtrace** Git repository.
4. Configure settings:
 - **Framework Preset:** Vite
 - **Root Directory:** client
 - **Build Command:** vite build
 - **Output Directory:** dist
5. Click **Deploy**. Copy your Vercel URL (e.g., <https://flowtrace.vercel.app>).

3. Deploy Express Backend on Render (Time: 7 Mins)

Render runs the Express server as a live Web Service.

1. Sign up/log in at **Render.com**.
2. Click **New + > Web Service**.
3. Connect your Git repository.
4. Enter configuration details:
 - **Name:** flowtrace-backend
 - **Runtime:** Node
 - **Build Command:** pnpm install && pnpm run build
 - **Start Command:** node dist/server/index.js
5. In the **Environment** variables section, add the following:

```
PORT=3001
MONGODB_DB=flowtrace
MONGODB_URI=mongodb://brijeshearn12_db_user:I63m3tiG03baEwnR@ac-shzqujs-shard-00-00.9rgipwa.mongodb.net:27017,ac-shzqujs-shard-00-01.9rgipwa.mongodb.net:27017,ac-shzqujs-shard-00-02.9rgipwa.mongodb.net:27017/flowtrace?ssl=true&replicaSet=atlas-54kbkk-shard-0&authSource=admin
CLIENT_URL=https://[YOUR-VERCEL-FRONTEND-URL]
LLM_ENABLED=false
```

6. Click **Create Web Service**. Copy your Render Backend API URL (e.g., <https://flowtrace-backend.onrender.com>).

4. Link Frontend to Backend (Time: 5 Mins)

1. In your local frontend files, ensure client API calls target your Render API URL (<https://flowtrace-backend.onrender.com>) instead of `localhost:3001`.
2. Push the changes to GitHub. Vercel will automatically redeploy the updated frontend in the background.

Congratulations! Your full-stack workflow builder is live and securely connected.